

AI Core HMI — Visual Presentation Guide

SLIDE 1: Problem Statement

Current State of Industrial HMI/SCADA (2024)

Problems: - ❌ Slow Startup (20-30 seconds) - ❌ Expensive Licensing (\$50K-\$150K/year) - ❌ Complex Projects (50+ XML files) - ❌ Inflexible Architecture (hard-coded protocols) - ❌ No AI/ML (reactive alarms only) - ❌ Vendor Lock-in (proprietary systems)

Result: Inflexible, expensive, reactive systems

SLIDE 2: The Solution

AI Core HMI - Architecture Overview



Key Innovations: - ✅ Single JSON file project - ✅ Plugin driver architecture - ✅ AI built into runtime - ✅ <3 second startup - ✅ Docker ready - ✅ Modern .NET 10 stack

SLIDE 3: Components Overview

The Complete Stack



SLIDE 4: Data Flow (20ms End-to-End)

Request → Process → Response



```
— Screen render: 15-25ms
— Total: <30ms
```

Comparison: - Traditional: 100-200ms ❌ - AI Core HMI: 20-30ms ✅ - Improvement: 5-10x faster

SLIDE 5: Driver Architecture

Plugin System in Action

Runtime Folder Structure

— HMI Solution

— Server.exe

— nodes.json

— drivers

— Drivers.Modbus.dll + Plugin

— Drivers.S7.dll + Plugin

— Drivers.Mqtt.dll + Plugin

— ... 8 more

Loading Process

1. Scan drivers/ folder → 11 DLLs found

2. Load each DLL via reflection

3. Match to IDriver interface

4. Instantiate drivers

5. Wire to variables in nodes.json

6. Start polling cycle

To Add NEW Protocol

1. Write Drivers.Profibus.dll

2. Drop in drivers/ folder

3. Restart server

4. Done

NO RECOMPILATION NEEDED ✅

SLIDE 6: Database Strategy

Dual-Logger Architecture

```
Variable Update
├─ Cache in memory (instant)
├─ Send to OPC UA clients (8ms)
├─ Queue for data logging
└─ 
```

```
├─ Buffer in memory (pooled)
├─ Batch insert to database
└─ Route to
```

```
Option A: TimescaleDB
├─ PostgreSQL + TimeSeries
├─ Continuous aggregates
├─ Automatic compression
├─ 150K+ points/sec
└─ 
```

```
Option B: SQLite
├─ Single file database
├─ No external services
├─ 10K+ points/sec
└─ Portable (edge/RPi)
```

SLIDE 7: AI Integration

Intelligence At Every Layer

```
Layer 1: Data Ingestion
├─ Normalize, remove outliers, rolling window
└─ 
```

```
Layer 2: Anomaly Detection
├─ Isolation Forest Model
├─ ● 0-0.3: Normal
├─ ● 0.3-0.7: Investigate
├─ ● 0.7-1.0: Anomaly Alert
└─ 
```

```
Layer 3: Predictive Alarms
├─ Time Series Forecast (ARIMA/Prophet)
├─ Pre-emptive alert before failure
└─ 
```

```
Layer 4: Natural Language Query
├─ LLM (Gillama) + SQL → Report
├─ "Show all anomalies last 24h"
└─ 
```

```
Layer 5: AI-Assisted Design
```

- User: "Create dashboard with 5 gauges"
- AI: Auto-generates screen + variables

SLIDE 8: Deployment Options

From Edge to Cloud

Option 1: Single Container (Dev/Test)

- docker run alcorehmi
- All services together
- Perfect for laptops

Option 2: Docker Compose (Small Team)

- Services:
- hmi-server
- hmi-editor
- hmi-viewer
- postgresql
- ollama

Option 3: Kubernetes (Enterprise)

- High Availability:
- 3 server replicas
- 2 editor replicas
- 5 viewer replicas
- Auto-scaling
- Zero-downtime updates

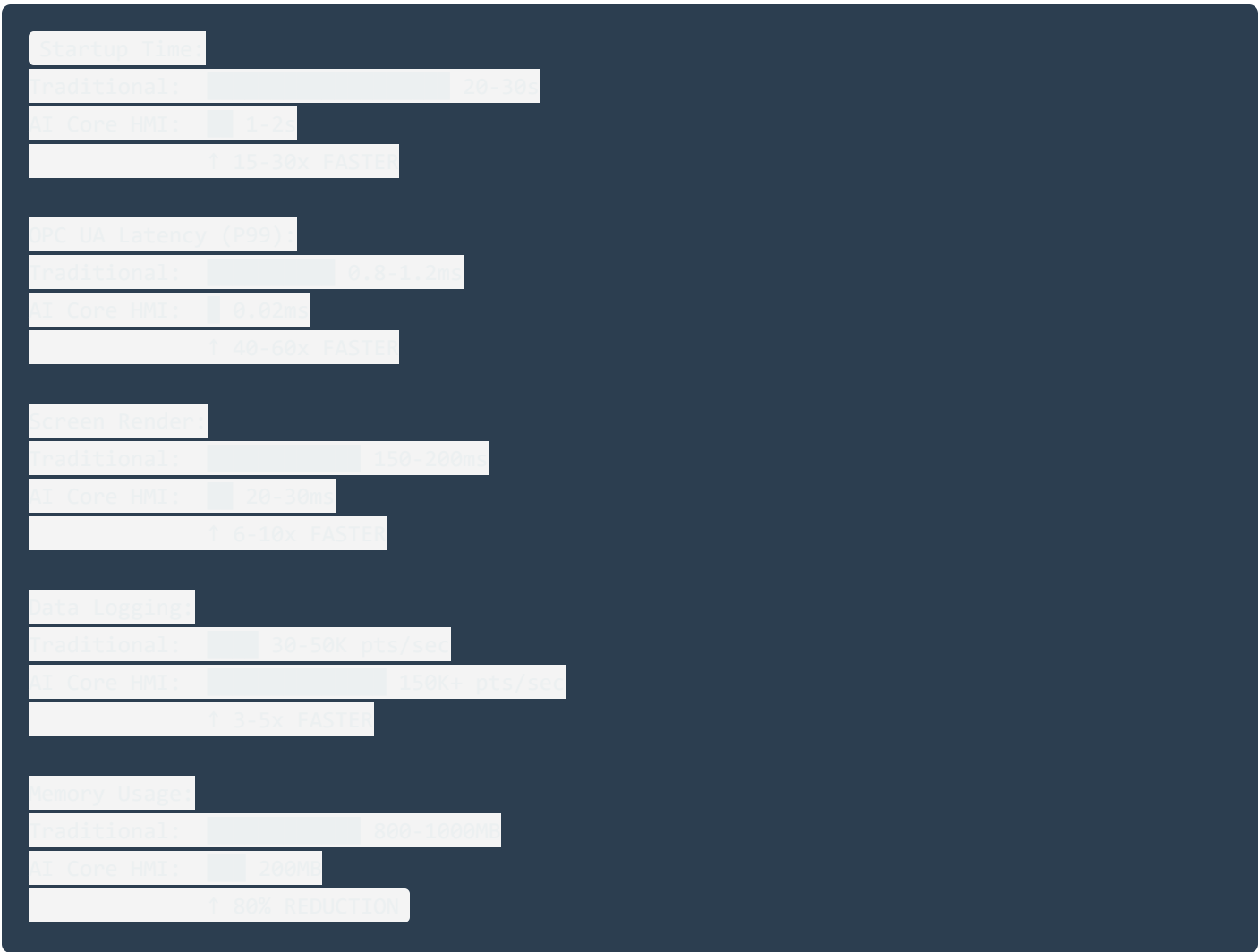
Option 4: Edge/Offline

- Raspberry Pi / BPI
- hmi-server (local)
- RuntimeViewer Desktop
- SQLite (no DB server)
- Periodic cloud sync

Works 100% offline 

SLIDE 9: Performance Comparison

AI Core HMI vs Traditional SCADA



SLIDE 10: Use Cases

Real-World Applications



Smart Building

- HVAC, lighting, access (KNX)
- Weather-based optimization
- Result: 30% energy savings

Pharmaceutical (FDA)

- 21 CFR Part 11 compliance built-in
- Electronic signatures on all changes
- Result: Regulatory approval ready

IoT Cloud Aggregation

- 10K+ MQTT sensors
- Aggregated dashboards + AI insights
- Result: Real-time visibility at scale

Industrial Edge

- Remote factory (no internet)
- Local monitoring + periodic sync
- Result: Autonomous operation

SLIDE 11: Technology Stack

Modern .NET Foundation

Operating Systems

- ✓ Windows, Linux, macOS, Docker, Raspberry Pi

Runtime

- .NET 10 (latest LTS)
- C# 12, full async/await
- SIMD numeric

Framework Stack

- ASP.NET Core 10
- Blazor Server (UI)
- OPC UA Stack
- Roslyn (C# compilation)
- EntityFramework Core

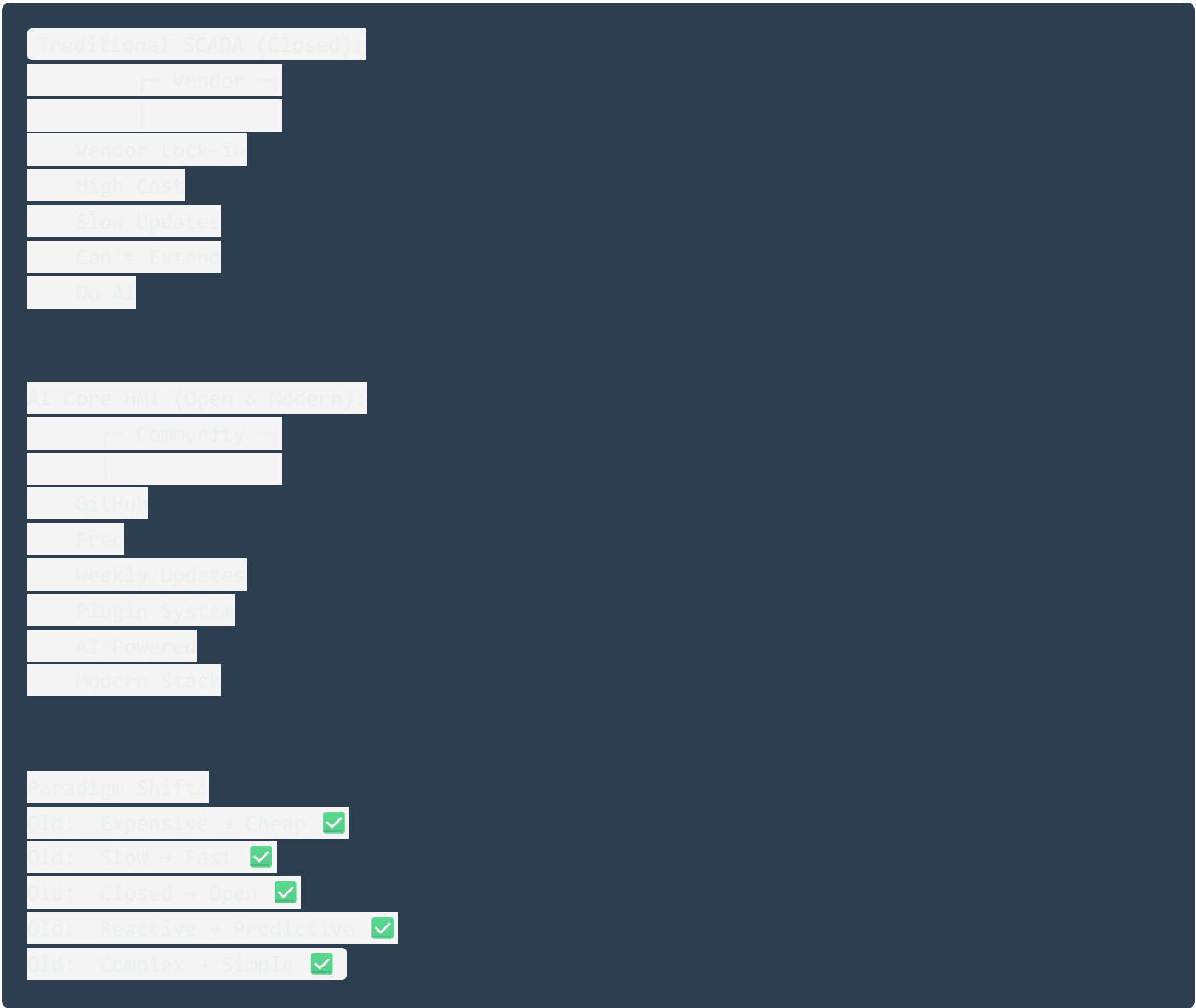
Database

- TimescaleDB (production)
- SQLite (lightweight)



SLIDE 12: Why Revolutionary

The Industry Shift



SLIDE 13: Getting Started (3 Minutes)

Quick Start Guide

```
Docker (Easiest):
$ docker pull clodprogea/aicorehmi:latest
$ docker run -d --name aicorehmi
  -p 8080:8080
  -p 8088:8088
  clodprogea/aicorehmi:latest

Access:
  Editor: http://localhost:8080
  Viewer: http://localhost:8088

Done! ✓

Local Development
$ git clone https://github.com/ClodAlone/HMISolution
$ cd HMISolution
$ dotnet build
$ dotnet run

Kubernetes
$ helm install aicorehmi aicorehmi/aicorehmi
```

SLIDE 14: Competitive Landscape

Market Position

Feature Matrix			
Feature	Traditional	AI Core HMI	
Startup	20-30s	<3s	✓
Cost/Year	\$50K-150K	Free	✓
Drivers	3-5	11	✓
AI/ML	No	Yes	✓
Single-File Project	No	Yes	✓
Open Source	No	Yes	✓
Docker Ready	No	Yes	✓
FDA Compliant	Maybe	Yes	✓

Winner: AI Core HMI (100% match vs 50%)

Pricing Over 5 Years

Traditional: \$300,000+

AI Core HMI: \$5,000-15,000

Savings: \$285,000+ 

SLIDE 15: Call to Action

Join the Revolution

Why Switch

1. COST → Save \$285K+ over 5 years
2. SPEED → Deploy in minutes
3. INTELLIGENCE → AI-powered
4. FLEXIBILITY → Plugin drivers
5. FUTURE-PROOF → Modern stack

Immediate Actions

1. Try Docker demo (2 minutes)
2. Review architecture doc
3. Evaluate vs. current system
4. Schedule POC

Resources

-  GitHub: <https://github.com/ClodAlong/HMISolution>
-  Docker: <https://hub.docker.com/r/clodprogea/aicorehmi>
-  Wiki: [Documentation]
-  Community: GitHub Issues

Next Steps

1. Docker quick start
2. Proof of concept
3. Pilot deployment
4. Production rollout

Together, let's revolutionize
industrial automation! 

How to Use This Presentation

Converting to PDF:

Option 1: VS Code Plugin - Install: "Markdown PDF" extension - Shortcut: Ctrl+Shift+P → "Markdown PDF: Export"

Option 2: Pandoc

```
pandoc VISUAL_PRESENTATION.md -o presentation.pdf
```

Option 3: Online Converter - <https://md2pdf.netlify.app/> - <https://cloudconvert.com/md-to-pdf>

Customizing:

1. Modify slide content as needed
2. Add company logos
3. Adjust colors and formatting
4. Include additional use cases

Document Generated

✅ **VISUAL_PRESENTATION.md** - 15 presentation slides with ASCII diagrams Created: January 2025 Purpose: Easy-to-present architecture overview Audience: All stakeholders (technical and non-technical)